

EDUCATION

Bachelor of Computer Science (Honors), Minor in Mathematics

- Awards: 4.0/4.0 GPA, Entrance Scholarships, Dean's Honor List

Sep. 2022 - Apr. 2027

Canada

EXPERIENCE

Amazon Web Services

Software Development Engineer Intern

- Returning Software Development Engineer Intern on the Amazon Aurora Limitless Database team

May 2026 – Jul. 2026

Canada

Quora

Machine Learning Engineer Intern

- Building new LLM & traditional ML-based features on the Poe.com platform team

Jan. 2026 – May 2026

Canada

Amazon Web Services

Software Development Engineer Intern

- Developed a TypeScript-based MCP server as a unified internal knowledge access point, enabling LLM agents to accurately answer developer queries by navigating documentation and tickets
- Built and evaluated a Retrieval-Augmented Generation (RAG) system capable of leveraging historical feedback, improving document retrieval speed by 50%+ and accuracy by 30%+ compared to baseline search APIs
- Engineered an ingestion pipeline to sync over 30,000 documents on internal wiki and Quip for the knowledge base weekly, including a filtering and labeling mechanism to exclude low-quality documents

May 2025 – Jul. 2025

Canada

Software Engineer Intern

- Designed and implemented Java Spring Boot microservice REST APIs of [redacted]
- Reduced database and server load with caching and the Factory pattern, improving API performance by 20%+
- Implemented a zero-disruption data storage logic migration from Cassandra to Postgres using a dual-write strategy
- Integrated ClickHouse into a Kafka-based data pipeline, enabling real-time dashboarding with 85% lower latency
- Refactored React UI components to enhance re-usability, reducing development time for new features

Sep. 2024 – Apr. 2025

Canada

Undergraduate Teaching Assistant

- Delivered mini-lectures and weekly tutorials on CS fundamentals, advanced algorithms, and Unix-like OS topics

Sep. 2023 – Apr. 2024; Sep. 2025 – Dec. 2025

Canada

PROJECTS

In-Place WebAssembly Interpreter | Rust, WebAssembly

[GitHub](#)

- Created a lightweight Wasm interpreter in Rust with full core spec compliance, based on research by Ben Titzer
- Improved CoreMark score by 71% vs. initial build via empirical cold-path tuning and cache-friendly optimizations
- Identified a Rust compiler miscompilation bug while building the project, contributed fix via a [merged pull request](#)

Kafkaesque | Elixir, TypeScript, gRPC, Phoenix, Svelte, Docker

[GitHub](#)

- Created a Kafka-like event streaming platform leveraging BEAM's fault-tolerance, supporting gRPC and REST APIs
- Implemented partitioned, demand-driven message processing with backpressure and retry logic using GenStage
- Developed observability features using Telemetry and Phoenix LiveView, including a real-time metrics dashboard

Outlier-Aware Image Classifier | Python, PyTorch, TensorFlow

[GitHub](#)

- Designed a novel image classifier with outlier-detection mechanism to minimize misuse and misclassification risk
- Devised custom loss, pretraining, and fine-tuning to enable outlier-robustness while improving in-domain accuracy

TECHNICAL SKILLS

Frameworks & Libraries: AWS SDK, AWS CDK, Spring Boot, JUnit, Node.js, Next.js, React, FastAPI, PyTorch, NumPy

Tools & Databases: Git, Docker, Kubernetes, Postman, Jira, Kafka, Cassandra, PostgreSQL, ClickHouse, DynamoDB

Languages: Java, Python, Rust, TypeScript, JavaScript, Go, C, C++, Elixir, Bash, OCaml, Haskell, Ruby, SQL